Zhengyang (Max) Wang

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Education

University of Michigan — Ann Arbor, MI
Bachelor of Science in Honors Mathematics, Honors Data Science — *Aug. 2024 – May 2027 (expected)*

- **GPA: 3.98/4.00**
- **Coursework**: Continuous Optimization*, Probability Theory*, Stochastic Processes*, Numerical Linear Algebra*, Machine Learning*, Data Structures & Algorithms, Regression Analysis* (* graduate-level)

University of California, Santa Barbara — Santa Barbara, CA
Major in Financial Mathematics and Statistics — *Sep. 2023 – Jun 2024*

- **GPA: 4.00/4.00**
- **Coursework**: Linear Algebra, Ordinary Differential Equations, Probability & Statistics, Data Science with R

Research Interests

Constrained Optimization, Vehicle Routing Problems, Game Theory

Research Experience

AKWA Fuel Dispatch Optimization — Mar 2026 - Present
MIT Data Science Lab Research Intern (Supervisor: Renfei Tan, Prof. David Simchi-Levi) — *MIT, IDSS*

- Developed a mixed-integer optimization framework in Python with HiGHS for fuel dispatch planning across 700+ service stations, modeling truck compartments, depot eligibility, product compatibility, and fleet type.
- Evaluated solver performance on real dispatch data, achieving up to 25% reduction in travel distance while maintaining approximately 96% of manual delivered volume on average across benchmark runs.
- Developed an LLM-assisted solution repair workflow that lets dispatchers express manual adjustments or temporary constraints in natural language and receive rerouting recommendations; to be presented at INFORMS 2026.

Multi-Mediator Routing Games — Jan 2026 - Present
Undergraduate Researcher (Supervisor: Prof. Achilleas Anastasopoulos) — *UMich, EECS*

- Developed a nonatomic routing-game model with multiple strategic mediators, private route recommendations, and ϵ-obedient user incentives.
- Defined fragmented obedient equilibrium and proved an existence result under lower-hemicontinuity assumptions.
- Built Braess-network experiments showing mediator fragmentation can generate equilibria between Wardrop and the social optimum.

Black Box Inverse Solver — Jan 2026 - May 2026
Research Assistant (Supervisor: Yidan Xu, Prof. Yixin Wang) — *UMich, Dept. of Statistics*

- Developed a meta-learning framework for uncertainty quantification in PDE inverse problems, mapping sparse spatiotemporal observations to calibrated joint confidence intervals over PDE parameters.
- Conducted ablation studies comparing Transformer, Set Transformer, and Set2Seq architectures, achieving calibration within 0.33 percentage points of nominal 95% joint coverage across PDE families.
- Co-authored a manuscript based on this work, submitted to NeurIPS 2026 for review.

Clinical NLP and Privacy-Preserving Methods — Jun 2025 - Jan 2026
Research Assistant (Supervisor: Dr. Siqi Li) — *Duke-NUS Medical School (virtual)*

- Developed clinical NLP pipelines on the MIMIC-IV dataset, transforming unstructured notes into patient-level embeddings using CUI co-occurrence matrices for downstream predictive modeling.
- Reviewed Privacy-Enhancing Technologies (PETs) in omics research, extracting standardized study variables to assess methodological rigor and reproducibility of privacy-preserving machine learning approaches.

Dynamic Pricing under Nonlinear Demand — Dec 2024 - Apr 2025
Research Assistant (Supervisor: Daniele Bracale, Prof. Moulinath Banerjee) — *UMich, Dept. of Statistics*

- Modeled competitive online pricing as a repeated seller-buyer game, where sellers learn unknown demand parameters from posted prices and observed purchase outcomes.
- Implemented Python simulations using online gradient descent and negative log-likelihood updates to study regret, price convergence, and learning behavior under nonlinear demand models.

Projects

Benchmarking Optimization Methods & Trust-Region Tolerance Strategies | *Python, Git* Apr 2026

- Benchmarked gradient descent, Newton, quasi-Newton, and trust-region methods on 12 unconstrained optimization problems spanning convex, ill-conditioned, smooth nonlinear, and nonconvex settings.
- Analyzed the tradeoff between convergence to stationary points and computational runtime, finding that L-BFGS delivered the strongest practical balance across problem types.
- Studied trust-region CG subproblem accuracy and showed that adaptive tolerance avoids manual tuning while preserving convergence behavior and runtime performance.

Adam Revisited: Momentum, Adaptivity, and Convergence | *Python, PyTorch, Git* Nov 2025

- Implemented SGD, AdaGrad, RMSProp, Adam, and AMSGrad from scratch, validating AMSGrad's restored theoretical convergence guarantees.
- Analyzed Adam vs. AMSGrad across non-stationary disaster-risk datasets and built visual diagnostics illustrating step-size behavior, second-moment dynamics, and convergence stability.

Mini AlphaGo | *Python, PyTorch, Git* Jan 2025 – Apr 2025

- Developed and optimized game-state preprocessing pipelines, converting raw Go board data into PyTorch tensors.
- Implemented a dual-head Residual Network (policy + value) for move prediction and win-rate estimation.
- Engineered a Monte Carlo Tree Search (MCTS) framework integrating neural priors and value backpropagation, realizing AlphaGo-style decision-making through self-play reinforcement loops.
- Project Repository: **github.com/MichiganDataScienceTeam/W25-mini-alphago**

Random Forests and Condorcet's Jury Theorem (*Supervisor: Sam Cochran*) | *Python* Oct 2024 - Dec 2024

- Investigated Random Forests via Condorcet's Jury Theorem and applied them to early-onset diabetes prediction.
- Selected speaker at the Michigan Math Club on Random Forests and their applications in healthcare prediction.

Work Experience

Math Learning Center Tutor Sep 2025 - Present
UMich, Dept. of Mathematics *Ann Arbor, MI*

- Provide 9+ hours/week of walk-in tutoring for Calculus I & II, supporting students with varied backgrounds.
- Lead weekly study groups of 5–10 students to build mathematical intuition and prepare for course exams.

Wealth Management Summer Intern Jul 2025 - Aug 2025
Huatai Securities *Nanjing, China*

- Produced 100+ market briefs and 150+ pages of presentations, integrating supply-chain analysis and product-level benchmarking across themes including BCI, quantum computing, nuclear fusion, and healthcare AI.
- Delivered daily client briefings across 1,000+ retail accounts, communicating market developments, evaluating risks and opportunities, and supporting strategic asset-allocation recommendations.

Technical Skills

Languages: C++, Python, Java, R, SQL